Mobility, Algorithms, and Attention: A Stochastic Adaptive Multiplex Model of Opinion Dynamics with Brownian Agents

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Abstract

We introduce a continuous-time model of opinion dynamics in which agents diffuse through physical space by Brownian motion while simultaneously interacting through an adaptive, algorithmically curated digital network. Opinions evolve under bounded-confidence attraction in the physical layer and under an engagement-weighted influence term in the digital layer, whose topology coevolves with the opinions through a platform recommendation kernel. The model incorporates a finite attention budget that forces digital interaction to displace physical interaction, an assimilation–indifference–repulsion influence function, and heavy-tailed influence heterogeneity. Numerical simulations across the four nested levels of the model yield four main results. (i) Physical mobility controls how many opinion clusters survive: the number of coexisting clusters decays with the diffusion coefficient D, with the crossover set by the competition between the local convergence time and the diffusive neighbourhood-renewal time. (ii) Under opinion-independent (Brownian) mobility, geography never becomes a predictor of opinion: spatial–opinion correlations remain at noise level for all D, because opposite-minded agents interleave in space instead of segregating. Ideological structure therefore lives in the network of interactions, not on the map. (iii) A modest amount of opinion-neutral long-range digital exposure is enough to melt the locally frozen state and recover the well-mixed mean-field cluster structure, but an adaptive similarity-driven recommendation algorithm reverses this effect: as algorithmic homophily increases, the digital layer reorganises into ideologically homogeneous, spatially delocalised echo chambers with near-perfect opinion assortativity and vanishing visible disagreement. (iv) Once influence includes a repulsive response to strongly opposed views, every platform design radicalises the population, and the ranking inverts: the opinion-neutral platform—the consensus machine of the pure bounded-confidence world—produces the fastest and most extreme polarization, because uncurated exposure maximises contact with repulsion-triggering content, while algorithmic homophily, by hiding disagreement, paradoxically dampens extremism. A phase diagram in the mobility–digital-attention plane (D, λ) locates the consensus, fragmentation, and polarization regimes.

1 Introduction

Continuous-opinion models with bounded confidence, introduced by Deffuant et al. [1] and Hegselmann and Krause [2], describe populations of agents who average their opinions only with peers whose views are already sufficiently close. Despite their simplicity, these models produce the three canonical outcomes of social influence—consensus, polarization, and fragmentation—and have become a standard laboratory for studying collective opinion formation [3–5].

Two ingredients of real societies are, however, usually absent from this laboratory. First, people move. Physical encounters are structured by geography, and geography changes as individuals commute, migrate, and mix; the consequences of mobility for bounded-confidence dynamics have only recently begun to be explored systematically [6]. Second, people increasingly interact through platforms whose connection topology is not given but *manufactured*: recommendation algorithms decide who sees whom, they do so as a function of the very opinions they help shape, and the resulting feedback between opinions and network structure is a coevolutionary dynamic in its own right [7–10]. Empirically motivated modelling of echo chambers points to the same loop of homophilic exposure and reinforcing influence [11], while multiplex formulations show that the coexistence of several interaction layers can qualitatively change consensus thresholds [12].

This paper combines these ingredients in a single stochastic model, built in four nested levels so that each mechanism can be switched on and its marginal effect isolated:

1. **Level 1.** Agents perform Brownian motion in a two-dimensional periodic domain and interact through a short-range spatial kernel with bounded confidence.
2. **Level 2.** A static, opinion-neutral digital layer of long-range links is added, sharing a finite attention budget with the physical layer.
3. **Level 3.** The digital layer becomes adaptive: a recommendation kernel rewires attention slots towards ideologically similar sources, with an algorithmic-homophily parameter γ.
4. **Level 4.** The platform is generalised to an arbitrary engagement function (similarity-driven, neutral, or controversy-driven), the influence function acquires an assimilation–indifference–repulsion structure [13, 14], and influence strengths become heavy-tailed, allowing influencer-like agents; stubborn "zealot" agents [15] are supported as special nodes.

The Brownian formulation makes the physical layer a genuine statistical-physics object: the box size L and the diffusion coefficient D define a spatial mixing time $\tau_{\text{mix}} \sim L^2/D$, which competes with the opinion convergence time $\tau_{\text{op}} \sim \alpha^{-1}$. Our central question is how this competition interacts with the digital layer: *can strong digital connectivity destroy the geographical structure of opinion while simultaneously increasing ideological polarization?*

Our simulations answer with a refinement that we did not anticipate when formulating the model. Mobility does control the surviving number of opinion clusters, as expected. But under opinion-independent mobility the geographical structure of opinion that digital media could "destroy" never forms in the first place: spatially close agents converge with their compatible neighbours, yet incompatible agents remain interleaved at the same locations, so position and opinion stay uncorrelated at every mobility. Geographic opinion structure, in this model class, requires opinion–position coupling (homophilic migration or segregation), not mere locality of interaction. What the digital layer does control—strongly—is the *network* structure of opinion: adaptive recommendation produces near-perfectly assortative, spatially delocalised ideological communities. A second surprise appears when influence can be repulsive: cross-cutting exposure of any kind then becomes the engine of radicalisation, and the platform designs reorder, with the neutral, uncurated platform driving the population to the extremes faster than either curated design.

2 Model

2.1 State variables

Each of N agents carries a position $\mathbf{r}_i(t) \in [0, L)^2$ with periodic boundary conditions and an opinion $x_i(t) \in [-1, 1]$. Agent i may also carry an influence strength s_i and a stubbornness flag. The digital layer is a directed graph $A_{ij}(t) \in \{0, 1\}$, where $A_{ij} = 1$ means that the platform currently shows content by agent j to agent i. Each agent has a fixed number k of *attention slots*, so that $\sum_j A_{ij} = k$ for all i and all t: attention is conserved, and new digital connections can only be acquired by dropping old ones.

2.2 Dynamics

Positions follow independent Brownian motions,

$$\mathrm{d}\mathbf{r}_i = \sqrt{2D}\,\mathrm{d}\mathbf{W}_i(t), \tag{1}$$

with diffusion coefficient D. Opinions obey the Itô equation

$$\mathrm{d}x_i = \alpha_i \frac{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)\,(x_j - x_i)}{\sum_j K(r_{ij})\, B_\epsilon(x_j - x_i)}\,\mathrm{d}t \;+\; \beta_i \frac{\sum_j A_{ij}\, s_j\, F(x_i, x_j)}{\sum_j A_{ij}\, s_j}\,\mathrm{d}t \;+\; \sigma_x\,\mathrm{d}B_i(t), \tag{2}$$

clipped to $[-1, 1]$, where r_{ij} is the minimum-image distance between agents i and j. The physical interaction kernel is a truncated Gaussian,

$$K(r) = e^{-r^2/2\ell^2}\,\mathbf{1}(r < 3\ell), \tag{3}$$

and $B_\epsilon(\Delta) = \mathbf{1}(|\Delta| < \epsilon)$ implements bounded confidence. The truncation in Eq. (3) is not cosmetic: because the drift is normalised, an untruncated Gaussian would let an agent with no nearby compatible peers average with the entire population at $O(1)$ rate through the exponential tail, making spatial isolation impossible and silently removing the mobility scale from the problem.

The influence function in the digital term has the assimilation–indifference–repulsion form

$$F(x_i, x_j) = \begin{cases} x_j - x_i, & |x_j - x_i| < \epsilon_1, \\ 0, & \epsilon_1 \le |x_j - x_i| < \epsilon_2, \\ -\eta\,(x_j - x_i), & |x_j - x_i| \ge \epsilon_2, \end{cases} \tag{4}$$

with the repulsive branch ($\eta > 0$) active only at Level 4; at Levels 2–3, F reduces to plain bounded confidence with threshold $\epsilon_1 = \epsilon$. Empirical support for repulsive ("negative") influence is discussed by Tang et al. [14].

2.3 Attention budget

The rates in Eq. (2) satisfy

$$\alpha_i = \alpha_{\mathrm{tot}}\,(1 - \lambda), \qquad \beta_i = \alpha_{\mathrm{tot}}\,\lambda, \qquad \lambda \in [0, 1], \tag{5}$$

so the digital attention share λ measures how much of a finite information-processing capacity is devoted to the platform. Increasing digital exposure therefore *displaces* physical influence rather than adding to it, together with the hard slot constraint $\sum_j A_{ij} = k$.

2.4 The platform: engagement-driven rewiring

The digital graph coevolves with the opinions. In each interval dt, agent i replaces one uniformly chosen attention slot with probability $\rho\,\mathrm{d}t$; the platform selects the replacement source j with probability

$$P(i \to j) \;\propto\; E(|x_i - x_j|)\; s_j, \tag{6}$$

where E is the platform's *engagement kernel*. We compare three platform designs:

$$E_{\mathrm{sim}}(\Delta) = e^{-\gamma\Delta}, \qquad E_{\mathrm{neu}}(\Delta) = 1, \qquad E_{\mathrm{con}}(\Delta) = e^{-(\Delta-\delta)^2/2s^2}. \tag{7}$$

The similarity-driven kernel models algorithmic homophily with strength γ [8, 9]; the neutral kernel is an opinion-blind baseline; the controversy-driven kernel encodes a platform that maximises engagement by showing users content at a preferred ideological distance δ—content different enough to provoke a reaction, a mechanism conceptually close to the post-distribution algorithms of de Arruda et al. [10].

2.5 Influence heterogeneity and zealots

At Level 4, influence strengths are drawn from a shifted Pareto distribution with tail exponent κ, normalised to unit mean, so a small number of agents ("influencers") dominate the recommendation distribution in Eq. (6) and the digital drift in Eq. (2). Optionally, a subset of agents can be declared stubborn ($\mathrm{d}x_i = 0$), representing institutional actors or media sources [15].

2.6 Observables

We measure: the polarization $P = \mathrm{Var}(x)$; the number of opinion clusters n_c (groups of at least two agents separated by gaps larger than 0.05 in sorted opinion space); a categorical state (consensus: $n_c \le 1$ or $P < 10^{-2}$; polarization: $n_c = 2$; fragmentation: $n_c \ge 3$); the spatial–opinion correlation, quantified both by Moran's I with Gaussian weights and by the *local agreement gap* $\Pr(|x_i - x_j| < \epsilon \mid r_{ij} < 3\ell) - \Pr(|x_i - x_j| < \epsilon)$; the opinion assortativity ρ of the digital graph (Pearson correlation of endpoint opinions across directed edges); the modularity Q of the digital graph with respect to the partition induced by opinion clusters; and the mean local disagreement $\langle |x_i - \bar{x}_{\partial i}| \rangle$ between an agent and the average opinion it is shown.

2.7 Numerical integration

Equations (1)–(2) are integrated by the Euler–Maruyama scheme. Unless stated otherwise, $N = 200$, $L = 1$, $\mathrm{d}t = 0.02$, $\alpha_{\mathrm{tot}} = 1$, $\epsilon = 0.3$, $\ell = 0.02$, $k = 10$, $\rho = 5$, and runs last $T = 60$ time units ($T = 80$ at Level 4); statistics are averaged over 6–8 independent realisations (3 for the phase diagram). With $\ell = 0.02$ the mean physical contact degree is $N\pi(3\ell)^2 \approx 2.3$, below the percolation threshold of the random geometric graph, so the instantaneous physical contact network is spatially fragmented and locality is meaningful. The complete implementation is available in the accompanying repository.

3 Results

3.1 Level 1: mobility sets the number of surviving clusters

Figures 1 and 2 summarise the purely physical model. The relevant comparison is between the local convergence time $\tau_{\mathrm{op}} \sim \alpha_{\mathrm{tot}}^{-1}$ and the time $\tau_\ell \sim \ell^2/D$ over which diffusion renews an agent's

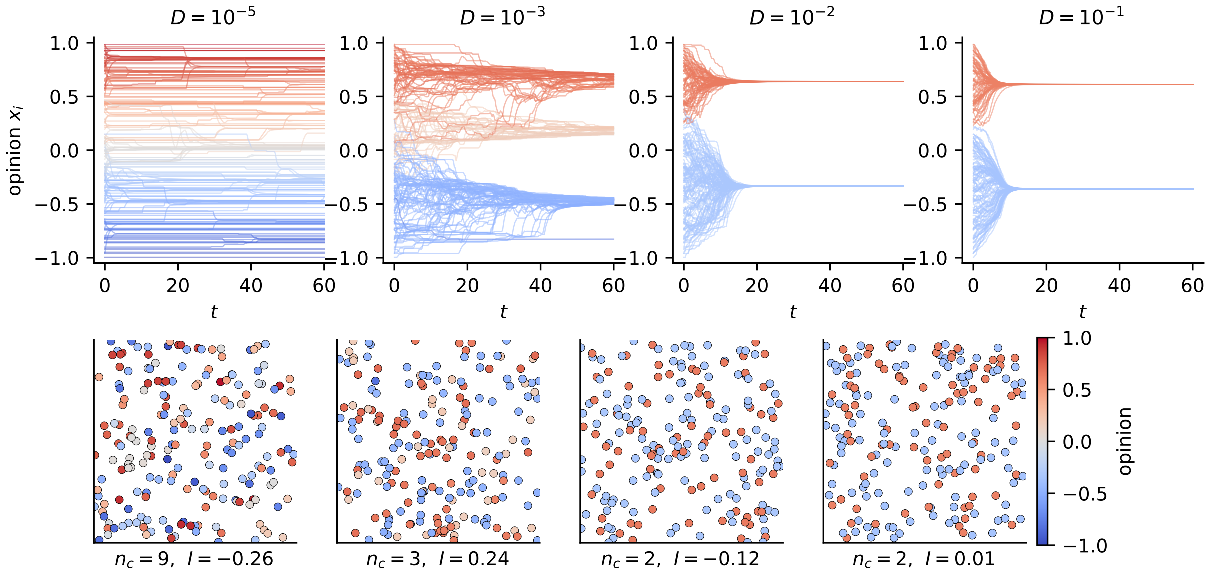

Figure 1: Purely physical dynamics (Level 1). Top: opinion trajectories $x_i(t)$ for increasing diffusion coefficient D; colours encode final opinions. Bottom: final spatial configurations; markers are agents coloured by opinion. Panel captions report the cluster count n_c and Moran's I. Low mobility freezes many coexisting clusters; high mobility approaches the mean-field bounded-confidence outcome. Note the absence of spatial colour domains at any D.

neighbourhood. For $D \lesssim 10^{-4}$ ($\tau_\ell \gg \tau_{\text{op}}$), compatible agents converge within their frozen local components and the population fragments into many coexisting clusters ($n_c \approx 5$–9). As D grows, neighbourhood renewal lets clusters within the confidence bound find each other and merge, and for $D \gtrsim 10^{-2}$ the system reproduces the well-mixed bounded-confidence outcome ($n_c \approx 2$–3 for $\epsilon = 0.3$) [1, 3]. This mobility-driven fragmentation-to-consensus crossover is consistent with the mobile-agent simulations of Alraddadi et al. [6].

The second panel of Fig. 2 contains the negative result announced in the Introduction. Moran's I—and, equivalently, the local agreement gap (not shown; it remains below 0.05 at all mobilities)—never departs from noise level, *even in the frozen regime where interactions are strictly local*. The reason is structural: Brownian positions are opinion-independent, so agents that disagree simply coexist at the same locations, interacting with disjoint compatible subsets. Locality controls *how many* opinions survive, but without opinion-dependent motion (homophilic relocation, avoidance, segregation) it cannot make geography informative about *which* opinion an agent holds. Claims that low mobility alone produces geographic echo chambers therefore need qualification: some coupling between opinion and position is a necessary ingredient, not a redundant detail.

3.2 Level 2: opinion-neutral long-range links melt the frozen state

Level 2 adds a static, uniformly random directed digital graph ($k = 10$ sources per agent) at low mobility, where the purely physical dynamics fragments into $n_c \approx 7$ clusters. Figure 3 shows the effect of shifting attention to this opinion-neutral layer. Because digital sources are sampled uniformly, each agent is exposed to the full global opinion distribution; digital neighbourhoods overlap globally, and a modest attention share ($\lambda \approx 0.2$) already suffices to merge the locally frozen clusters into the mean-field cluster set of the well-mixed bounded-confidence dynamics ($n_c \approx 3$

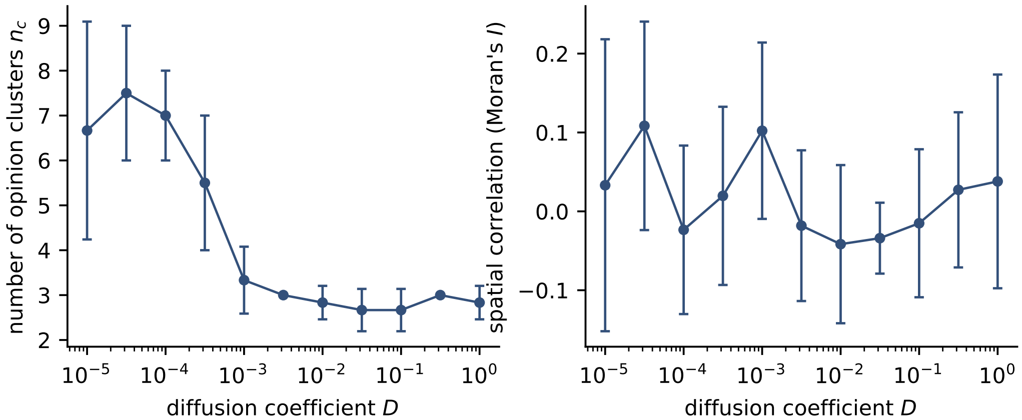

Figure 2: Level-1 sweep over the diffusion coefficient. Left: mean number of surviving opinion clusters. Right: Moran's I between opinion and position (mean $\pm$ s.d. over 6 realisations). Mobility interpolates between local fragmentation and mean-field behaviour, while the spatial–opinion correlation stays at noise level throughout.

at $\epsilon = 0.3$), beyond which additional digital attention changes nothing. Opinion-blind long-range exposure thus acts exactly like infinite-range mobility: it removes the excess fragmentation caused by locality—though full consensus would require a larger confidence bound, as in the classical model [1, 12]. This makes the contrast with the next level sharp: the consequences of connectivity depend entirely on how it is curated.

3.3 Level 3: algorithmic homophily builds delocalised echo chambers

At Level 3 the platform rewires attention slots with the similarity kernel $E_{\rm sim}$. Figure 4 traces the closing of the feedback loop opinion $\to$ topology $\to$ exposure $\to$ opinion as γ increases. For $\gamma \approx 0$ the layer behaves like the neutral platform of Level 2. As γ grows, the digital graph reorganises around the emerging opinion clusters: assortativity climbs towards one, the modularity of the digital graph with respect to opinion communities rises, and the average disagreement an agent experiences in its feed collapses even though the population as a whole remains fragmented. This is an echo chamber in the precise sense of Baumann et al. [11]: high global variance, near-zero locally visible disagreement. The mechanism mirrors the algorithmic-bias model of Sîrbu et al. [8]—biased exposure freezes fragmentation that neutral exposure would have healed—and the link-recommendation feedback of Santos et al. [9], transplanted into a coevolving multiplex [7]. Crucially, these ideological communities are *spatially delocalised*: their members are scattered uniformly across the box, so the polarised society shows no geographic signature whatsoever.

3.4 Level 4: engagement-maximising platforms and repulsion

Level 4 activates the repulsive branch of Eq. (4) and heavy-tailed influence strengths, and compares the three platform designs of Eq. (7) (Fig. 5). The outcome (Fig. 6) is unambiguous on one point: once influence can be repulsive, *every* platform design radicalises the population. All three designs end with two antagonistic blocs near the boundaries of opinion space. But the designs differ sharply in degree and speed, and the ordering is the opposite of what the pure bounded-confidence levels would suggest. The neutral platform is the most polarizing ($\mathrm{Var}(x) \approx 0.98$, opinions pinned

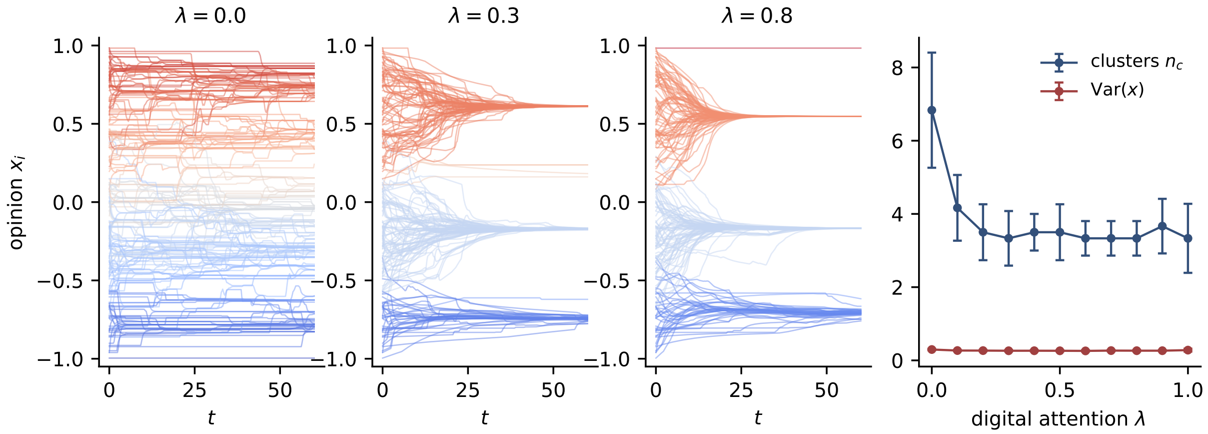

Figure 3: Level 2: static random digital layer at low mobility ($D = 10^{-4}$). Left three panels: opinion trajectories for increasing digital attention share λ. Right: cluster count and polarization versus λ (mean $\pm$ s.d.).

at ± 1 within a few time units): uncurated exposure samples the full opinion distribution, which at the start is dominated by pairs beyond the repulsion threshold ϵ_2, so almost every recommendation triggers mutual repulsion. The controversy-driven platform is nearly as extreme ($\mathrm{Var}(x) \approx 0.89$) but slower, because its engagement peak at $\delta = 0.8 < \epsilon_2$ places a substantial fraction of exposures inside the indifference band where they do nothing. The similarity-driven platform is the *least* polarizing ($\mathrm{Var}(x) \approx 0.68$): by hiding strongly opposed content it starves the repulsive channel, and the drift to the extremes—driven by the residual cross-cutting links that stochastic rewiring keeps producing—is slow and incomplete. The network signatures also decouple from the opinion signatures: the neutral platform reaches maximal polarization with *zero* opinion assortativity ($\rho \approx 0$), i.e. polarization without echo chambers, whereas both curated platforms end almost perfectly assortative ($\rho \approx 0.9$–1.0).

Two lessons follow. First, exposure to disagreement, often proposed as the remedy for echo chambers, here *is* the engine of radicalisation, as anticipated by models of partial antagonism [10, 13] and consistent with the empirical evidence that negative influence is the better-supported micro-mechanism [14]. Second, whether a given platform design polarises is not a property of the design alone but of its interaction with the psychology of influence: the neutral platform is the consensus machine of the assimilation-only world (Sec. 3.2) and the radicalisation machine of the repulsive world, while homophilic curation switches roles in the opposite direction. In every case the society is simultaneously highly connected (every feed remains full) and deeply divided: connectivity and polarization are not opposites.

3.5 Phase diagram: mobility versus digital attention

Figure 7 assembles the global picture in the (D, λ) plane for the full Level-4 model. The horizontal axis interpolates between a purely geographical society and a purely platform-mediated one; the vertical axis spans four decades of mobility. At $\lambda \approx 0$ the physical layer decides the outcome and the Level-1 crossover is recovered: frozen local fragmentation at low D, the mean-field cluster set at high D. As λ grows, the platform captures the attention budget, the repulsive channel activates through the residual cross-cutting exposure, and both polarization and extremism rise

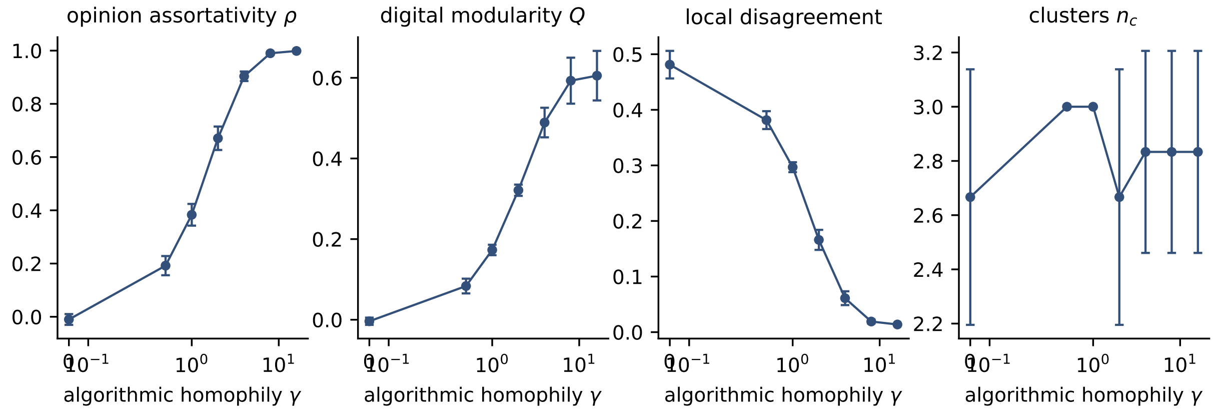

Figure 4: Level 3: adaptive digital layer with similarity-driven recommendation, $\lambda = 0.5$, $D = 10^{-3}$. From left to right: opinion assortativity of the digital graph, modularity with respect to opinion clusters, mean local disagreement, and cluster count as functions of the algorithmic homophily γ (mean $\pm$ s.d. over 6 runs).

across *all* mobilities: the high-λ region is a bimodal, radicalised state essentially independent of D. Digital attention thus neutralises the moderating effect of physical mixing—once the platform curates most of an agent's exposure, it no longer matters how fast people move. Geography sets the outcome only for societies whose attention it still owns.

4 Discussion

Summary. We built a stochastic adaptive multiplex model coupling Brownian spatial motion, bounded-confidence social influence, and an algorithmically curated digital layer under a finite attention budget. The model reproduces, within one framework, the mobility-controlled fragmentation crossover of mobile-agent models [6], the defragmenting effect of opinion-neutral long-range coupling [12], the echo-chamber feedback of algorithmic homophily [8, 9, 11], and repulsion-driven radicalisation [13], and it locates these outcomes in a single (D, λ) phase diagram.

The geography null result. The most instructive finding is the one we did not design for. In the hypothesis motivating this work, strong digital connectivity would *destroy* geographical opinion structure while increasing ideological polarization. The simulations show that under opinion-independent mobility there is no geographical opinion structure to destroy: locality shapes how many opinions survive but leaves position and opinion uncorrelated, because disagreeing agents interleave rather than segregate. The empirically observed geographic clustering of opinion must then arise from mechanisms outside this model class—homophilic migration, socially structured mobility, or spatially correlated exogenous influence. This sharpens the modelling agenda: adding opinion-dependent drift to Eq. (1) (a Schelling-type term) is the natural Level 5, and the present model provides the null baseline against which its effect can be measured cleanly.

No platform design is uniformly safe. Levels 2–4 give a graded answer to the public debate on whether platforms polarise, and the answer is conditional on the micro-psychology of influence. If influence is purely assimilative (bounded confidence), the neutral platform is actively beneficial—

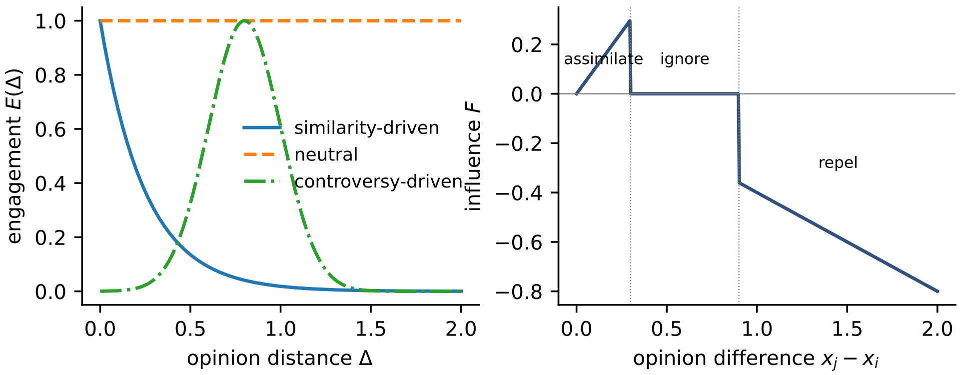

Figure 5: Level-4 ingredients. Left: the three engagement kernels of Eq. (7) ($\gamma = 4$, $\delta = 0.8$, $s = 0.2$). Right: the assimilation–indifference–repulsion influence function of Eq. (4) ($\epsilon_1 = 0.3$, $\epsilon_2 = 0.9$, $\eta = 0.4$).

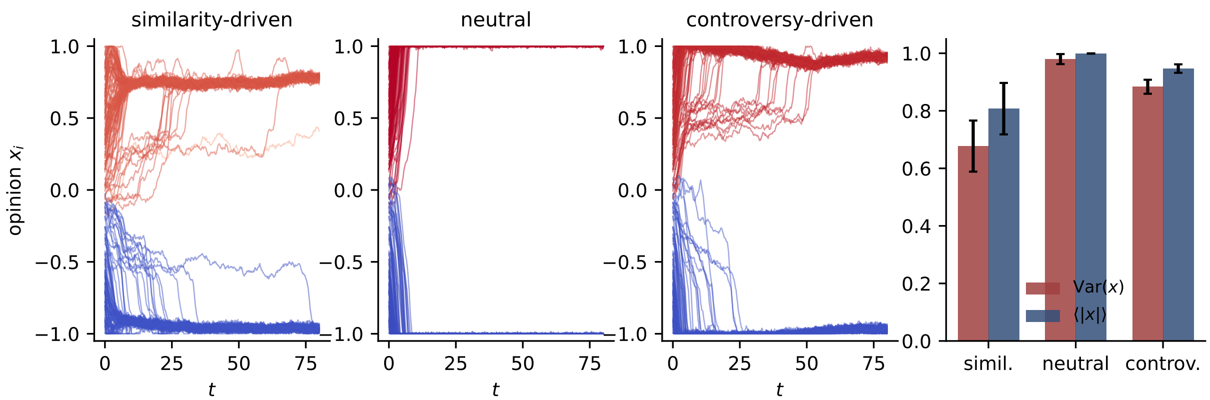

Figure 6: Level 4: platform designs compared under repulsive influence and heavy-tailed influence strengths ($\lambda = 0.6$). Left three panels: representative opinion trajectories under similarity-driven, neutral, and controversy-driven engagement. Right: polarization $\mathrm{Var}(x)$ and extremism $\langle|x|\rangle$ (mean $\pm$ s.d. over 8 runs).

it supplies the long-range mixing that geography denies to immobile populations—and the harm comes from similarity curation, which freezes fragmentation and hides it from its participants behind feeds with vanishing visible disagreement. If influence has a repulsive component, the roles invert: uncurated and controversy-seeking exposure manufacture extremism, and homophilic curation, whatever its epistemic vices, acts as a buffer against radicalisation. Interventions on recommendation algorithms that ignore which influence regime the population is in can therefore backfire in either direction—increasing cross-cutting exposure is precisely the wrong medicine when negative influence dominates, which is the mechanism with the stronger empirical support [14].

Limitations and outlook. Opinions here are one-dimensional and confidence bounds static; adaptive-confidence extensions are natural [16]. The attention budget is a fixed split rather than a dynamic allocation, engagement is a function of opinion distance only, and stubborn agents were

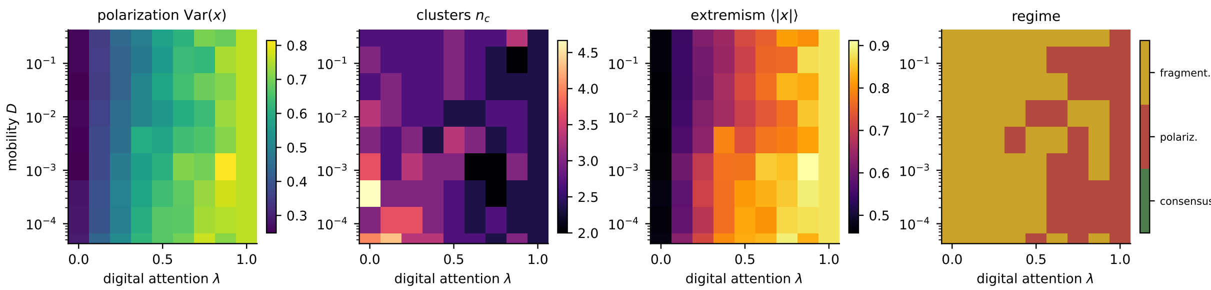

Figure 7: Phase diagram of the full Level-4 model (similarity-driven engagement, repulsion, heavy-tailed influence) in the mobility–digital-attention plane. From left to right: polarization $\mathrm{Var}(x)$, cluster count n_c, extremism $\langle|x|\rangle$, and the majority categorical state over 3 realisations per grid point.

implemented but not systematically explored—sweeping their number and placement against the platform designs is an obvious next step [15]. Analytically, the Brownian formulation invites a continuum (McKean–Vlasov) limit in which the physical term becomes a nonlocal interaction on a density field and the mixing-time competition can be made precise; the phase boundaries of Fig. 7 are the natural target of such a calculation [4, 5].

Reproducibility

All code, figures, and animation scripts are contained in the accompanying repository. Each figure is produced by a single script under `scripts/`; the model implementation lives in `src/socialsim/`; Manim animations of the coupled dynamics are in `visualizations/`.

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